

Disaster Relief Management System

Overview

The Disaster Relief Management System (DRMS) is a Python and MySQL based application developed to assist in organizing and managing disaster relief operations. The system helps maintain records of donors, victims, volunteers, resources, and relief centers, enabling efficient coordination during emergency situations.

This project demonstrates the integration of Python with MySQL to perform database operations and manage real-world data through a command-line interface.

Features

Donation Management

- Register donors and record donations.
- Store donor information securely in the database.
- Track donation amounts and donor locations.

Victim Assistance Management

- Register victims affected by disasters.
- Record assistance requirements and current status.
- Maintain area-wise victim records.

Volunteer Management

- Register volunteers for relief operations.
- Store volunteer contact information, roles, and assigned relief centers.

Resource Management

- View available resources and inventory information.
- Support resource tracking during disaster response.

Relief Center Directory

- Maintain information about relief centers.
- Access relief center records quickly when required.

Database Integration

- Uses MySQL for efficient storage and retrieval of records.
 - Supports structured data management through SQL queries.
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Technologies Used

- Python
 - MySQL
 - mysql-connector-python
 - SQL
 - Command Line Interface (CLI)
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Database Tables

The project utilizes the following database tables:

- DONOR
 - VICTIMS
 - VOLUNTEERS
 - RESOURCES
 - R_CENTERS
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Skills Demonstrated

- Python Programming
 - Database Management
 - SQL Query Execution
 - CRUD Operations
 - Exception Handling
 - Input Validation
 - Software Design
 - Problem Solving
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Project Workflow

1. Establish connection with MySQL database.
2. Display the main menu.
3. Allow users to:
4. Make donations
5. Request assistance
6. Register as volunteers
7. View donor records
8. View volunteer records
9. View resources
10. View relief center information
11. Store and retrieve data using MySQL queries.
12. Close the database connection safely.

Future Improvements

- Graphical User Interface (GUI)
- Web-Based Dashboard
- User Authentication System
- Real-Time Resource Tracking
- Automated Report Generation
- Emergency Notification System
- Data Visualization and Analytics

Author

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Project Status

Completed as an academic project and continuously being improved to enhance functionality and user experience.